Examination practice: solutions

2A — 30 October 2026

The solutions show the equation, the substitution and the answer. Each exercise ends with the common mistakes and the wrong numbers they produce: use them to check your own work. $g = 10$ N/kg.

A Speed and distance–time graphs

Exercise 1 — The round trip

Solution — **(a)** Less than 75 km/h. The driver spends more time at the lower speed (60 km/h), so the average is pulled towards 60 km/h: average speed is total distance over total time, not the mean of the speeds.

$$\textbf{(b)}\quad t_1 = \frac{s}{v} = \frac{60}{90} = 0.667 \text{ h (40 min)}, \qquad t_2 = \frac{60}{60} = 1.00 \text{ h}$$

$$v_{\text{av}} = \frac{s_{\text{total}}}{t_{\text{total}}} = \frac{120}{1.667} = 72 \text{ km/h}$$

$$\textbf{(c)}\quad v = \frac{72}{3.6} = 20 \text{ m/s}$$

Check and common mistakes — (a) The mean 75 km/h is the most common wrong answer. (b) Writing 40 min as 0.40 h gives $120/1.40 = 86$ km/h. Using only one way (60 km) gives $60/1.667 = 36$ km/h. (c) Multiplying by 3.6 gives 259 m/s, which is not possible for a car. Check: $20 \times 3.6 = 72$.

Exercise 2 — Overtaking

Solution —

$$\textbf{(a)}\quad v_B = \frac{180}{3.0} = 60 \text{ km/h}$$

$$v_A = \frac{240 - 0}{3.0 - 1.0} = 120 \text{ km/h}$$

(b) The two lines cross: both cars are at the same distance from P at the same time. They cross at $t = 2.0$ h, $d = 120$ km. After this point the line of A is above the line of B: A is ahead.

$$\textbf{(c)}\quad d_B(1.5) = 60 \times 1.5 = 90 \text{ km}, \qquad d_A(1.5) = 120 \times (1.5 - 1.0) = 60 \text{ km}$$

$$\Delta d = 90 - 60 = 30 \text{ km}$$

(d) A reaches B when the two distances are equal.

$$120\,(t - 1.5) = 60\,t \;\Rightarrow\; 60\,t = 180 \;\Rightarrow\; t = 3.0 \text{ h}$$

Here the lines meet at the very end of the graph (180 km).

Check and common mistakes — (a) Reading the speed of A as $240/3.0 = 80$ km/h forgets that A starts at 1.0 h. (b) Giving the time as 1.0 h (where A starts) or the distance in hours. (c) Subtracting the times (not the distances). (d) Taking $t = 1.5 + 1.5 = 3.0$ by luck: check with the equation, not with the picture.

Exercise 3 — Timing a trolley

Solution — **(a)** Run 4 (4.3 s): it is far from the others. Likely cause: late start or early stop of the stopwatch (human reaction time), or the trolley was released badly.

$$\textbf{(b)} \quad t_{\text{mean}} = \frac{3.0 + 2.9 + 3.1 + 3.0}{4} = 3.0 \text{ s}$$

$$\textbf{(c)} \quad v = \frac{s}{t} = \frac{1.50}{3.0} = 0.50 \text{ m/s}$$

(d) Systematic error: every time is shorter than the true time by the same amount. A time that is too small gives a speed that is too large ($1.50/2.8 = 0.54$ m/s).

(e) Use light gates or a video, or a longer distance (a longer time makes the 0.2 s a smaller fraction).

Check and common mistakes — (b) Including 4.3 s gives 3.26 s and $v = 0.46$ m/s. (d) Calling it a random error: random errors change from run to run, here every run changes the same way. (d) Concluding that the speed is too small.

B Conservation of energy

Exercise 4 — The loop

Solution — **(a)** Released from 0.80 m the trolley reaches the top of the loop with all its energy as gravitational potential energy: $E_k = 0$, $v = 0$. It needs at least 2.0 m/s there, so it needs kinetic energy at the top as well.

(b) Energy at release = energy at the top (no friction):

$$mgh = mgH + \tfrac{1}{2}mv^2 \;\Rightarrow\; h = H + \frac{v^2}{2g}$$

(c)

$$h = 0.80 + \frac{(2.0)^2}{2(10)} = 0.80 + 0.20 = 1.0 \text{ m}$$

(d) Still 1.0 m: the mass cancels in (b).

(e)

$$\text{lowest point:} \quad \tfrac{1}{2}mv^2 = mgh \;\Rightarrow\; v^2 = 2(10)(1.5) = 30 \;\Rightarrow\; v = 5.5 \text{ m/s}$$

$$\text{top:} \quad v^2 = 2g(h - H) = 2(10)(1.5 - 0.80) = 14 \;\Rightarrow\; v = 3.7 \text{ m/s}$$

$3.7 > 2.0$ m/s: it stays on the track.

Check and common mistakes — (b) Writing $h = v^2/2g$ forgets the height of the top: gives 0.20 m. Forgetting g: $h = 0.80 + 4.0/2 = 2.8$ m. Forgetting the $\frac{1}{2}$: $0.80 + 0.40 = 1.2$ m. (e) At the top using h instead of $h - H$ gives $v = 5.5$ m/s: the trolley has already used part of its energy to climb 0.80 m.

Exercise 5 — A skater leaves the half-pipe at an angle

Solution — **(a)** Lower than A. At the highest point she is still moving, so she still has kinetic energy; the total energy is the same as at A, so less of it is gravitational potential energy.

$$\textbf{(b)} \quad \Delta E_p = mg\Delta h = (55)(10)(2.5 - 1.2) = 715 \text{ J}$$

$$\tfrac{1}{2}mv^2 = 715 \;\Rightarrow\; v^2 = \frac{2(715)}{55} = 26 \;\Rightarrow\; v = 5.1 \text{ m/s}$$

$$\textbf{(c)} \quad mgh_A = mgh_{\max} + \tfrac{1}{2}mv^2 \;\Rightarrow\; h_{\max} = h_A - \frac{v^2}{2g}$$

$$\textbf{(d)} \quad h_{\max} = 2.5 - \frac{(3.0)^2}{2(10)} = 2.5 - 0.45 = 2.05 \text{ m}$$

(e) The same, 2.05 m: the mass cancels.

Check and common mistakes — (a) Answering "the same height as A because energy is conserved": energy is conserved, but not all of it is potential at the top. (d) Giving 2.5 m. Adding instead of subtracting gives 2.95 m. Using $1.2 + 0.45 = 1.65$ m: the kinetic energy at the top is subtracted from the energy at A, not from E.

Exercise 6 — Pendulum with a peg

Solution —

$$\textbf{(a)} \quad \tfrac{1}{2}mv^2 = mgh \;\Rightarrow\; v^2 = 2(10)(0.80) = 16 \;\Rightarrow\; v = 4.0 \text{ m/s}$$

(b) 0.80 m. The energy at the lowest point is the same on both sides and the bob stops (instant) when all of it is gravitational potential energy again. The peg does no work: it changes the shape of the path (the string is shorter, the angle is bigger), not the height.

(c) Air resistance and friction at the pivot transfer some energy to the thermal store of the surroundings.

(d) None: the mass cancels in $\frac{1}{2}mv^2 = mgh$.

Check and common mistakes — (a) Using $h = 1.0$ m (the peg) gives 4.5 m/s; using the length 1.2 m gives 4.9 m/s. (b) Answering 1.0 m (the peg height) or 0.80 m "because the string is shorter, it rises more": the height depends only on the energy.

Exercise 7 — The track with two hills

Solution — **(a)** Hill 1 only. The ball has gravitational potential energy corresponding to 2.0 m: it cannot reach a point higher than 2.0 m. Hill 1 is 1.5 m, hill 2 is 2.4 m.

$$\textbf{(b)} \quad \tfrac{1}{2}mv^2 = mg(2.0-1.5) \;\Rightarrow\; v^2 = 2(10)(0.50) = 10 \;\Rightarrow\; v = 3.2 \text{ m/s}$$

(c) The ball climbs the left side of hill 2, slowing down. At a height of 2.0 m (the height of S) it stops for an instant and then rolls back down.

$$\textbf{(d)} \quad v^2 = 2g(2.0-0.30) = 2(10)(1.7) = 34 \;\Rightarrow\; v = 5.8 \text{ m/s}$$

(e) No. The ball still turns back at 2.0 m: the mass cancels and the height reached depends only on the release height.

Check and common mistakes — (b) Using 1.5 m (the height of the hill) gives $v = 5.5$ m/s. (c) Answering 2.4 m. (d) Using 0.30 m gives 2.4 m/s. (e) "A heavier ball has more energy so it passes": it has more kinetic energy but also more weight to lift.

Exercise 8 — Speed and height: from data to g

Solution — **(a)** $v^2 = 10,\ 20,\ 30,\ 40$ $(\text{m/s})^2$ (10.2, 20.3, 30.3, 39.7 before rounding).
(b) A straight line through the origin ($v^2 \propto h$).

$$\textbf{(c)} \quad mgh = \tfrac{1}{2}mv^2 \;\Rightarrow\; v^2 = 2g\,h \quad (\text{of the form } y = \text{gradient} \times x) \;\Rightarrow\; \text{gradient} = 2g$$

$$\textbf{(d)} \quad \text{gradient} = \frac{39.7}{2.00} = 19.8 \text{ (m/s)}^2\text{/m} \;\Rightarrow\; g = \frac{19.8}{2} = 9.9 \text{ N/kg}$$

$$\textbf{(e)} \quad v^2 = 19.8 \times 8.0 = 158 \;\Rightarrow\; v = 12.6 \text{ m/s}$$

Check: h is 4 times as big as 2.0 m $\Rightarrow$ v is 2 times as big: $2 \times 6.3 = 12.6$ m/s.

Check and common mistakes — (a) Writing v^2 with the units m/s. (c) Saying the gradient is g (forgetting the 2). (d) Dividing by 2 twice. (e) Multiplying $6.3 \times 4 = 25$ m/s: v is not proportional to h.

Exercise 9 — Air resistance: energy bookkeeping

Solution —

$$\textbf{(a)} \quad E_p(\text{top}) = mgh = (2.0)(10)(5.0) = 100 \text{ J}; \quad E_p(\text{midway}) = 50 \text{ J}; \quad E_p(\text{bottom}) = 0$$

$$\text{thermal (midway)} = 100 - 50 - 40 = 10 \text{ J}; \qquad \text{thermal (bottom)} = 100 - 0 - 75 = 25 \text{ J}$$

$$\textbf{(b)} \quad \tfrac{1}{2}mv^2 = 75 \;\Rightarrow\; v^2 = \frac{2(75)}{2.0} = 75 \;\Rightarrow\; v = 8.7 \text{ m/s}$$

$$\textbf{(c)} \quad \tfrac{1}{2}mv^2 = 100 \;\Rightarrow\; v = 10 \text{ m/s}$$

$$\textbf{(d)} \quad F_{\text{upper}} = \frac{10}{2.5} = 4.0 \text{ N}, \qquad F_{\text{lower}} = \frac{25-10}{2.5} = \frac{15}{2.5} = 6.0 \text{ N}$$

(e) The force of air resistance increases with speed, and the ball is faster in the lower half.

Check and common mistakes — (a) Writing 25 J as the thermal energy "in the lower half": the table is cumulative, the lower half is $25 - 10 = 15$ J. (d) Using 25 J over 2.5 m gives 10 N for the lower half. Using 5.0 m for each half. (b) $v = \sqrt{75}$ not 75.

C Doing work

Exercise 10 — A crane lifts a load

Solution —

(a) $W = Fd = (5000)(12) = 60\,000$ J, $\qquad \Delta E_p = mg\Delta h = (500)(10)(12) = 60\,000$ J

The two values are equal: all the work becomes gravitational potential energy.

(b) $W = (5600)(12) = 67\,200$ J

(c) $67\,200 - 60\,000 = 7200$ J $\rightarrow$ kinetic energy of the load

(d) $\frac{1}{2}mv^2 = 7200 \;\Rightarrow\; v^2 = \dfrac{2(7200)}{500} = 28.8 \;\Rightarrow\; v = 5.4$ m/s

Check and common mistakes — (a) Comparing a force with an energy. (c) "Thermal energy": nothing rubs here. (d) Forgetting the square root gives 28.8 m/s; forgetting the $\frac{1}{2}$ gives $v = 7.6$ m/s.

Exercise 11 — A ramp does not reduce the work

Solution —

(a) $W = mg\Delta h = (30)(10)(1.0) = 300$ J

(b) $F = \dfrac{W}{d} = \dfrac{300}{1.0} = 300$ N

(c) 300 J: the box ends with the same gain in gravitational potential energy and no kinetic energy; without friction the work done equals the energy gained.

(d) $F = \dfrac{W}{d} = \dfrac{300}{5.0} = 60$ N

(e) $F = \dfrac{300}{10} = 30$ N, $\qquad W = 300$ J (unchanged)

(f) Wrong: a ramp reduces the *force*, but the distance is bigger. The work is the same.

Check and common mistakes — (d) Using $F = mg = 300$ N for the ramp too, or dividing 300 by 1.0 instead of 5.0. (e) Saying the work is halved because the force is halved: the distance doubles. (f) "Less effort so less work" is the usual misconception.

Exercise 12 — Stopping distances

Solution —

$$\textbf{(a)} \quad E_k = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(1000)(20)^2 = 200\,000 \text{ J}$$

$$\textbf{(b)} \quad F = \frac{E_k}{d} = \frac{200\,000}{40} = 5000 \text{ N}$$

$$d = \frac{E_k}{F} \quad \text{(work done by the brakes = kinetic energy lost)}$$

(c) (i) E_k doubles, so $d = 80$ m. (ii) $E_k \propto v^2$ is 4 times bigger: $d = 4 \times 40 = 160$ m. (iii) $d = 200\,000/10\,000 = 20$ m. (iv) $E_k = \frac{1}{2}(500)(40)^2 = 400\,000$ J, so $d = 400\,000/5000 = 80$ m.

Check and common mistakes — (ii) Saying 80 m: the kinetic energy depends on the *square* of the speed. (iv) Multiplying the factors the wrong way: halved mass gives $\times\frac{1}{2}$, doubled speed gives $\times 4$: $\frac{1}{2} \times 4 = 2$, so $2 \times 40 = 80$ m.

Exercise 13 — A ball in the sand

Solution —

$$\textbf{(a)} \quad \tfrac{1}{2}mv^2 = mgh \;\Rightarrow\; v^2 = 2(10)(3.0) = 60 \;\Rightarrow\; v = 7.7 \text{ m/s}$$

$$\textbf{(b)} \quad E_k = mgh = (0.50)(10)(3.0) = 15 \text{ J}$$

$$\textbf{(c)} \quad E = E_k + mg\Delta h = 15 + (0.50)(10)(0.040) = 15.2 \text{ J}$$

$$\textbf{(d)} \quad F = \frac{E}{d} = \frac{15.2}{0.040} = 380 \text{ N}$$

$$\textbf{(e)} \quad F = \frac{15 + (0.50)(10)(0.080)}{0.080} = \frac{15.4}{0.080} \approx 190 \text{ N}$$

About half: for the same energy, twice the distance means half the force.

Check and common mistakes — (c) Forgetting the potential energy lost while sinking gives 15 J and $F = 375$ N: the small difference is why the check must be done. (d) Using the distance 3.0 m instead of 0.040 m gives $F = 5.1$ N. (d) Using the weight $mg = 5.0$ N as the force.

D Power and efficiency

Exercise 14 — Three cyclists

Solution —

(a) $W_{\text{Anna}} = mg\Delta h = (60)(10)(120) = 72\,000$ J, $W_{\text{Ben}} = (90)(10)(120) = 108\,000$ J, $W_{\text{Chen}} = 72\,000$ J

(b) $P = \dfrac{W}{t}$: $P_{\text{Anna}} = \dfrac{72\,000}{600} = 120$ W, $P_{\text{Ben}} = \dfrac{108\,000}{900} = 120$ W, $P_{\text{Chen}} = \dfrac{72\,000}{480} = 150$ W

(c) No. Ben took 1.5 times as long, but he also did 1.5 times as much work. W/t is the same: 120 W.

(d) Ben transferred the most energy (108 kJ); Chen had the greatest power (150 W).

Check and common mistakes — (a) Forgetting to convert minutes into seconds: $72\,000/10 = 7200$ W. (c) "More time, less power" is true only for the same work.

Exercise 15 — The chain of efficiencies

Solution —

(a) $E = 0.40 \times 500 = 200$ MJ

(b) $E = 0.90 \times 200 = 180$ MJ

(c) $E = 0.80 \times 180 = 144$ MJ

(d) $\text{efficiency} = \dfrac{E_{\text{useful}}}{E_{\text{total}}} = \dfrac{144}{500} = 0.288 \approx 29\%$ $\qquad (0.40 \times 0.90 \times 0.80)$

(e) The first stage alone wastes 60% of the energy. Each following stage can only waste more of what is left, so the overall efficiency cannot be more than 40%.

(f) $E_{\text{gas}} = \dfrac{144}{0.70} = 206$ MJ $<$ 500 MJ

The gas burner needs less than half the chemical energy.

Check and common mistakes — (d) The mean 70% is the usual wrong answer; adding the three wasted amounts $60 + 10 + 20 = 90\%$ gives 10%. (f) Multiplying $144 \times 0.70 = 101$ MJ instead of dividing.

Exercise 16 — Two pumps

Solution —

(a) $P_A = 0.60 \times 2000 = 1200$ W, $\qquad P_B = 0.90 \times 1500 = 1350$ W

(b) $E_A = Pt = (1200)(60) = 72\,000$ J, $\qquad E_B = (1350)(60) = 81\,000$ J

$$m = \frac{E}{g\Delta h}: \quad m_A = \frac{72\,000}{(10)(12)} = 600 \text{ kg}, \qquad m_B = \frac{81\,000}{120} = 675 \text{ kg}$$

(c) $E_{\text{wasted,A}} = (2000)(60) - 72\,000 = 48\,000$ J, $\qquad E_{\text{wasted,B}} = (1500)(60) - 81\,000 = 9000$ J

(d) Wrong: what counts is the *useful* power, input power multiplied by efficiency. B has the smaller input but the greater useful power (1350 W against 1200 W), so it lifts more water.

Check and common mistakes — (a) Taking the input power as the useful power. (b) Forgetting that the time is 60 s: $m = 1200/120 = 10$ kg. (c) Using the useful energy as the wasted energy.

E Energy resources

Exercise 17 — A hydroelectric power station

Solution —

(a) $E = mg\Delta h = (2.0 \times 10^3)(10)(50) = 1.0 \times 10^6$ J in 1 s $\Rightarrow$ $P = 1.0$ MW

(b) $P_{\text{out}} = 0.80 \times 1.0 = 0.80$ MW

(c) $N = \dfrac{4.0}{0.80} = 5$

(d) $P = 0.80 \times \dfrac{(2.0 \times 10^3)(10)(30)}{1} = 0.80 \times 6.0 \times 10^5 = 4.8 \times 10^5$ W $= 480$ kW

(e) Advantage: renewable, no fuel burnt, no CO_2 while generating. Disadvantage: the dam floods land and habitats; the output depends on the rainfall (dry season).

Check and common mistakes — (a) Giving the answer in J: it is a power, so W (J every second). (c) Using 1.0 MW (the input) gives 4 stations. (d) Not reducing the height gives 800 kW.

Exercise 18 — Choosing energy resources

Solution — **(a)** coal N; wind R; geothermal R; nuclear fission N; tidal R; biofuel R.
(b) Any two of: geothermal (heat from radioactive decay inside the Earth); nuclear fission (energy stored in the nuclei of uranium); tidal (gravitational interaction of the Earth with the Moon).
(c) Whether a resource is renewable depends on whether it can be replaced as fast as it is used, not on waste. Uranium is non-renewable because the supply of uranium on Earth is finite and cannot be replaced on a human time scale.
(d) Problem: the output is intermittent (no sun at night, no wind on calm days), so supply may not match demand. Ways to reduce it: store energy (batteries or pumped storage), use other sources to fill the gaps, connect electricity grids over a wide area.

Check and common mistakes — (a) Classifying nuclear as renewable "because it has no CO_2" mixes renewable with low carbon. (b) Attributing geothermal or tidal to the Sun. (d) Saying "solar cells are expensive": this is not a problem of supply.

F Paper 2 style: multiple choice

Exercise 19 — Multiple choice

Solution — Answers: 1-C, 2-A, 3-C, 4-B, 5-D, 6-C, 7-D, 8-C.

1. At the highest point only the vertical part of the velocity is zero; the horizontal motion goes on, so there is kinetic energy (A is the most common wrong answer).
2. $E_k = \frac{1}{2}mv^2 \to \frac{1}{2}(3m)(v/2)^2 = \frac{3}{4}\cdot\frac{1}{2}mv^2 = 0.75\,E_k$. B (1.5) multiplies by 3 and by $\frac{1}{2}$ forgetting the square; C (6) multiplies by 3 and by 2.
3. $P = W/t = Fd/t = (20)(5.0)/4.0 = 25$ W. A forgets the time (100 J), B multiplies force and time (80), D is d/t.
4. efficiency $= 200/800 = 25\%$. A is the wasted fraction (600/800); C is the useful energy divided by the wasted energy (200/600); D is the ratio turned upside down (800/200).
5. Renewable means that the resource is replaced as fast as it is used. Uranium is finite. A, B and C may be true, but they are not the reason.
6. Straight line through the origin: constant speed $= 20/4.0 = 5.0$ m/s at all times. A is 20/8; B doubles the speed because the time doubles; D is the distance at $t = 8$ s.
7. The bag does not move, so $d = 0$ and $W = Fd = 0$. B, C, A are numbers that multiply the weight, 100 N, by something.
8. $v^2 = 2gH$: v doubles when H is multiplied by 4.

G Challenge

Exercise 20 — The roller coaster with friction

Solution —

$$\textbf{(a)}\quad E_k = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(400)(18)^2 = 64\,800 \text{ J}$$

$$\textbf{(b)}\quad \Delta E_p = mg\Delta h = (400)(10)(20) = 80\,000 \text{ J}$$

$$\textbf{(c)}\quad E_{\text{thermal}} = 80\,000 - 64\,800 = 15\,200 \text{ J}$$

$$\textbf{(d)}\quad F = \frac{E_{\text{thermal}}}{d} = \frac{15\,200}{100} = 152 \text{ N}$$

$$\textbf{(e)}\quad \text{energy needed} = \Delta E_p + Fd = (400)(10)(15) + (152)(60) = 60\,000 + 9120 = 69\,120 \text{ J}$$

The car has only 64 800 J of kinetic energy at the lowest point: $64\,800 < 69\,120$, so the car does **not** reach the top of the second hill.
(f) Yes: without friction it would need only 60 000 J, and it has 64 800 J, so it would arrive with 4800 J of kinetic energy ($v \approx 4.9$ m/s).

Check and common mistakes — (d) Using the speed 18 m/s as if it were part of the force. (e) Comparing only the potential energy (60 000 J) with the kinetic energy (64 800 J) forgets the friction on the way up: the conclusion would be wrong. (e) Using 100 m of track instead of 60 m.